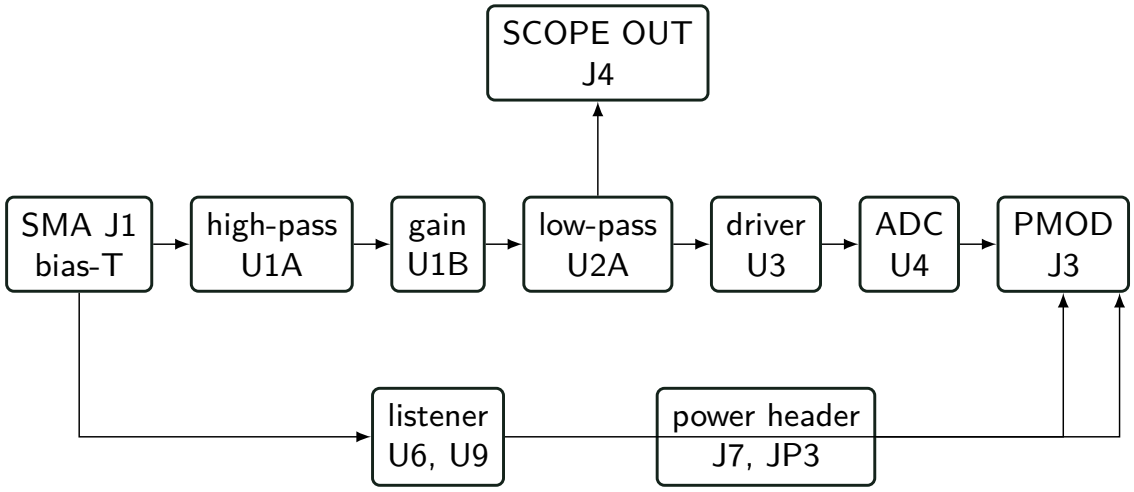
Slide 12: board and layout

A 125 × 21.5 mm strip that fits adjacent PMOD ports

ADS7057 test board v3 — study notes

1 The floor plan



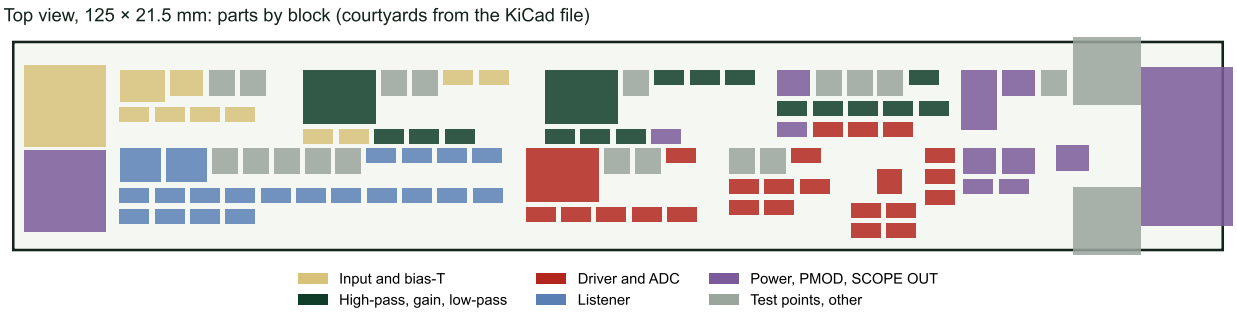
The signal runs from the SMA on the left to the PMOD on the right. The listener, the power links and SCOPE OUT sit around that line.

2 The board in numbers

Size and stack	125 × 21.5 mm, two copper layers, 1.6 mm FR-4
Parts	101 footprints: SMD parts, 4 through-hole connectors plus the 5-pin header, 2 mounting holes, 18 test points, 2 solder jumpers (JP1, JP3)
Tracks and vias	886 track segments (836 at 0.2 mm, 50 at 0.15 mm); 214 vias of 0.6 mm with a 0.3 mm drill, 119 of them ground stitching
Ground	a pour on both layers, tied by the stitching vias
Longest analog net	LPF_OUT, 39 mm (it feeds SCOPE OUT and the driver)

The width of 21.5 mm comes from the PMOD pitch: ports sit about 22.7 mm apart, so several boards fit side by side. A 5 mm track delays a signal by about 30 ps and a 500 kHz wave lasts 2 µs, so track length does not matter in the analog chain. What matters is where the capacitors sit.

3 Top view by block



Every part belongs to one block. The ADC corner is built tight: the two 10 Ω resistors, the four 100 pF capacitors, the supply capacitors and the three 22 Ω resistors all sit around U4. AINP and AINM are 28 and 27 mm long, and a 1 mm difference is harmless.

4 What the board offers you on the bench

- Test points after every stage: TP3 SIG_IN, TP4 HPF_OUT, TP5 GAIN_OUT, TP6 LPF_OUT, TP7 and TP8 driver outputs, TP9 and TP10 ADC inputs, TP11 +3V3, TP12 AVDD, TP13 VMID5, TP14 VOCM, TP15 LST, TP16 WAKE_N, TP17 BIG_N, TP18 REFL, TP19 and TP20 ground.
- J7, a 5-pin power header (Onboard +, Force +, Sense +, Sense −, Force −) for the PMOD supply or a source-measure unit. Solder jumpers JP1 (head power) and JP3 (capture rail). SCOPE OUT (J4) copies the signal the ADC sees through 51 Ω.

5 Cost (JLCPCB, economic assembly, shipped to Chile)

From the calibrated model in `hw/tools/cost.py` (real quote of 2026-10-12): two ADC boards $66.84 in goods, five $117.82; five heads $40.77. Two ADC boards and five heads with shipping and 19 % IVA: about $141 to $154. Sixteen cards plus sixteen heads would be about $383 in goods alone ($305 and $79), so a 16-channel system does not fit under 200 000 CLP (about $210).

> For 16 channels the same card plugs into a backplane (FMC adapter and two 8-slot modules, `hw/backplane/`). The card itself does not change.